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Prof. Paolo Favaro and the Postdoctoral Selection Committee
Computer Vision Group (CVG), Institute of Computer Science
University of Bern
Neubrückstrasse 10, CH-3012 Bern, Switzerland

Re: Application for Postdoctoral Position in Deep Learning (Collaborative World Models & Rapid Adaptation)

Dear Prof. Favaro and Members of the Selection Committee,

I am writing to express my enthusiastic application for the Postdoctoral Researcher positions in Deep Learning within the Computer Vision Group (CVG) at the University of Bern, conducted in collaboration with Prof. Andrea Vedaldi (University of Oxford). As a final-year PhD candidate in Computer Vision and Machine Learning at the University of Ghana (thesis defence expected in October 2026), my research focuses on the intersection of **inference-time coordination of frozen visual representations**, **geometric multi-view consensus**, and **process-level self-supervision for visual reasoning**. My doctoral work directly addresses the foundational questions underlying your research program: how autonomous visual models can maintain distinct representations, reconcile conflicting evidence, and rapidly adapt to complex physical environments without gradient-based parameter updates ($\Delta\theta = 0$).

Alignment with Collaborative World Models and Multi-Agent Visual Coordination

Your initiative on **Collaborative World Models**—investigating independently operating visual agents that maintain distinct memories, perspectives, and specializations while coordinating to solve complex visual tasks—resonates directly with my doctoral research. In my PhD thesis, *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations*, I developed **Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)**. Rather than forcing visual models into an early monolithic representation, RCASA treats diverse foundation models (such as 3D vision-language models and 2D promptable segmenters) as independent epistemic agents. The framework mathematically separates source-specific confidence from contextual reliability, coordinating their hypotheses via probabilistic calibration and non-parametric consensus. This experience provides an immediate mathematical and algorithmic foundation for modeling how collaborative visual models can communicate, negotiate, and align their internal spatial representations.

Mitigating Premature Semantic Collapse & Test-Time Adaptation

A major discovery in my doctoral research—formalized in my recent ICLR 2027 submission, *Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation*—is the characterisation of **premature semantic collapse** as a primary bottleneck in multi-model composition. Standard visual pipelines discard secondary hypotheses prematurely via hard top-1 selection (argmax). Through matched top-k retention interventions on the 312 scenes ($\sim$50M points) of ScanNet200 and ScanNet++, I demonstrated that preserving distributional semantic alternatives allows secondary hypotheses to be recovered through cross-model interaction, boosting segmentation harmonic mean substantially. This finding is directly applicable to world models and embodied AI: an agent predicting environmental evolution must maintain a distribution over plausible future states rather than collapsing to a deterministic single hypothesis, ensuring resilience during test-time adaptation under distribution shift.

Geometric Consistency and Label-Free Process Supervision

In recent work submitted to SACAIR 2026, I explored multi-view physical agreement as an intrinsic supervisory signal for multimodal reasoning. We developed a label-free process reward for **Qwen2.5-VL-7B** trained with online **GRPO/VERL** and **vLLM**, leveraging geometric consistency across viewpoints to supervise visual reasoning without human-annotated reward models. Through counterfactual stress-testing, we eliminated reward-hacking failure modes (reducing spatial area bias correlation from $\rho > 0.72$ to 0.334). This bridges generative perception with self-supervised reinforcement learning, directly supporting your group's goals in generative video modeling, memory systems, and embodied decision-making.

Technical Execution, Systems Rigor, and Collaborative Fit

I bring rigorous engineering skills in **PyTorch**, **CUDA/Triton** fundamentals, and distributed inference. I have authored papers accepted or under review at leading venues (ICLR 2027 submission, ACCV 2026, WACV 2027, SACAIR 2026, and a comprehensive theoretical paper on RCASA in preparation for IEEE TPAMI). Having founded a software consultancy and served as a university lecturer, I combine deep technical autonomy with proven mentorship and collaborative skills.

I would welcome the opportunity to contribute to the pioneering collaborative world models and rapid adaptation initiatives at the University of Bern and the University of Oxford. Thank you for your time and consideration.

Sincerely,

Silas Kwabla Gah
PhD Candidate, University of Ghana